

Case 48: WU-E uniform-community spatial convergence

1 Verification purpose

This case verifies spatial convergence of the WU-E ellipse–heat propagation model in a uniform idealized community. It holds the physical domain, initial burning area, wind, design-fire curve, linear characteristic building dimension, and separation fixed while refining the raster spacing through 20, 10, 5, and 2.5 m. The question is whether physically located ignition times, the community propagation rate, and domain-integrated received heat approach grid-independent values.

Reading the notation. WU-E denotes the Wildland–Urban ELMFIRE model. HRR is heat-release rate; HRRPUA is heat-release rate per unit area. DFC means direct flame contact, TOA means time of arrival, and ROS means rate of spread. FTP is the material fire-thermal property used in the intended heat-to-spread law.

Current result: FAIL. 4 of 4 required variants have evaluable evidence. Workflow: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The uniform-community context follows the published WU-E parametric study [1]. Executable semantics were checked against `elmfire_spread_rate.f90` and `elmfire_level_set.f90` at source revision `a2dfbcd7220`. Those mapped files were byte-identical to the revision; unrelated pre-existing edits elsewhere meant the checkout as a whole was dirty. The intended downscaled model interprets `BLDG_AREA` as the linear characteristic building dimension L_c in metres. For each spacing it uses

$$\beta_{ssd} = \min\left(1, \frac{s_{sd} + L_c}{\Delta x}\right), \quad g_f = \min\left[\left(\frac{\Delta x}{L_c}\right)^2, 1\right], \quad R_{search} = \left\lceil \frac{HAZ}{\Delta x} \right\rceil. \quad (1)$$

For a target cell receiving direct-flame-contact and radiative fluxes, the cell-step energy is

$$E = (q_{DFC} + q_{rad}) \Delta t \Delta x^2. \quad (2)$$

The material fire-thermal property is fixed, with nominal $F_{TP,crit} = 5400 \text{ kJ m}^{-2}$, and the absolute urban spread magnitude is

$$U = \frac{60E}{0.3048 \Delta t \Delta x F_{TP,crit}}. \quad (3)$$

WU-E then partitions this magnitude among head, flank, and backing directions using the wind- and geometry-dependent ellipse before ELMFIRE advances the level set with its two-stage integration. The pinned source omits β_{ssd} and g_f , uses a cell-count heat band, and replaces FTP using a 30,000-kJ branch; these are expected convergence defects, not oracle terms.

3 Derivation of expected behavior

Let Q_h denote any physical observable calculated on spacing h . A convergent discretization has

$$Q_h = Q + Ch^p + o(h^p), \quad p \approx \frac{\log(\|Q_{2h} - Q_h\|/\|Q_h - Q_{h/2}\|)}{\log 2}. \quad (4)$$

Arrival times are evaluated at fixed physical landmarks by interpolation of the cross-community median TOA profile; community ROS is $1/b$ from $t = a + bx$ over ignited downwind columns. Total received heat is the sum of ELMFIRE’s terminal DFC and radiation energy rasters, whose stored values already include $\Delta t \Delta x^2$.

The four prepared grids use $h = 20, 10, 5, 2.5$ m with one fixed $\Delta t = 0.25$ s, so temporal-model and integration effects are not deliberately co-refined with space. Feedback-level timestamps must also show the complete 1,200-step sequence at that configured step. For a scalar Q , the finest-pair change is

$$e_Q = \frac{|Q_5 - Q_{2.5}|}{\max(|Q_{2.5}|, 10^{-12})};$$

for the three-component landmark vector, absolute values are replaced by the Euclidean norm. The reported order uses the three finest levels,

$$p_Q = \log_2 \frac{\|Q_{10} - Q_5\|}{\|Q_5 - Q_{2.5}\|}.$$

Only an exactly or roundoff-unchanged three-level sequence satisfies the flat-sequence alternative and is reported as such rather than as an invented infinite order. A flat finest pair following a nonzero coarse change fails this order component as possible stagnation or quantization.

4 Verification metrics and acceptance criteria

All four current-fingerprint variants and their exact-stop TOA, total-DFC, and total-radiation rasters are mandatory. Each concrete namelist and prepared raster geometry must agree with the case specification and manifest. Each feedback log must contain every expected start time from 0 through $t_{stop} - \Delta t$, with maximum absolute error 0.051 s to accommodate the log’s 0.1 s print precision. Finest-pair relative changes must not exceed 5% for landmark TOA and community ROS or 10% for total heat. A resolvable nonzero sequence must have observed order at least 0.5; a roundoff-flat three-level sequence is the explicit alternative. Every TOA fit must have $R^2 \geq 0.95$.

These are preregistered numerical-resolution criteria, not tolerances quoted by the references. Five percent makes a propagation observable materially smaller than a one-cell coarse-grid displacement; the 10-percent heat limit is looser because source/receiver overlap changes discontinuously with rasterization. The positive order floor of 0.5 detects contraction without asserting a formal order for this coupled, clipped model, and $R^2 = 0.95$ prevents a visibly nonlinear or fragmentary TOA profile from being summarized by a single ROS. The limits are not adjusted to the result.

5 Simulation configuration and predicted outcome

The physical community is 160 m by 40 m; every interior cell is FBFM 91 and uses a constant 20 m linear characteristic building dimension, 10 m separation, and building fuel model 14. A 20 m-wide upwind strip is initially burning. The wind is 15 mph from the west, the design fire reaches 800 kW m^{-2} at 30 s, and the run ends at 300 s. Firebrands, crown fire, stochastic wind, and suppression are disabled. A point ignition scheduled after the stop time prevents the generic early-stall shortcut without affecting the solution.

Quantity	Values	Units	Control intent
Grid spacing	20, 10, 5, and 2.5 m	m	Ratio-two refinement
Timestep / stop	0.25 / 300	s	Isolates spatial refinement
Physical source	20 by 40	m	Equal initial burning area
Wind / direction	15 / 270	mph / degree from	Fixed ellipse forcing
WU-E band	Entire physical domain	cells	Equal heat-interaction support
Terminal evidence	TOA, total DFC, total radiation	s / kJ	Propagation and energy convergence

Figure 1 is generated directly from the prepared 5 m GeoTIFFs. It appears before any result figure so the fuel field and initial level-set support can be audited independently of simulation output.

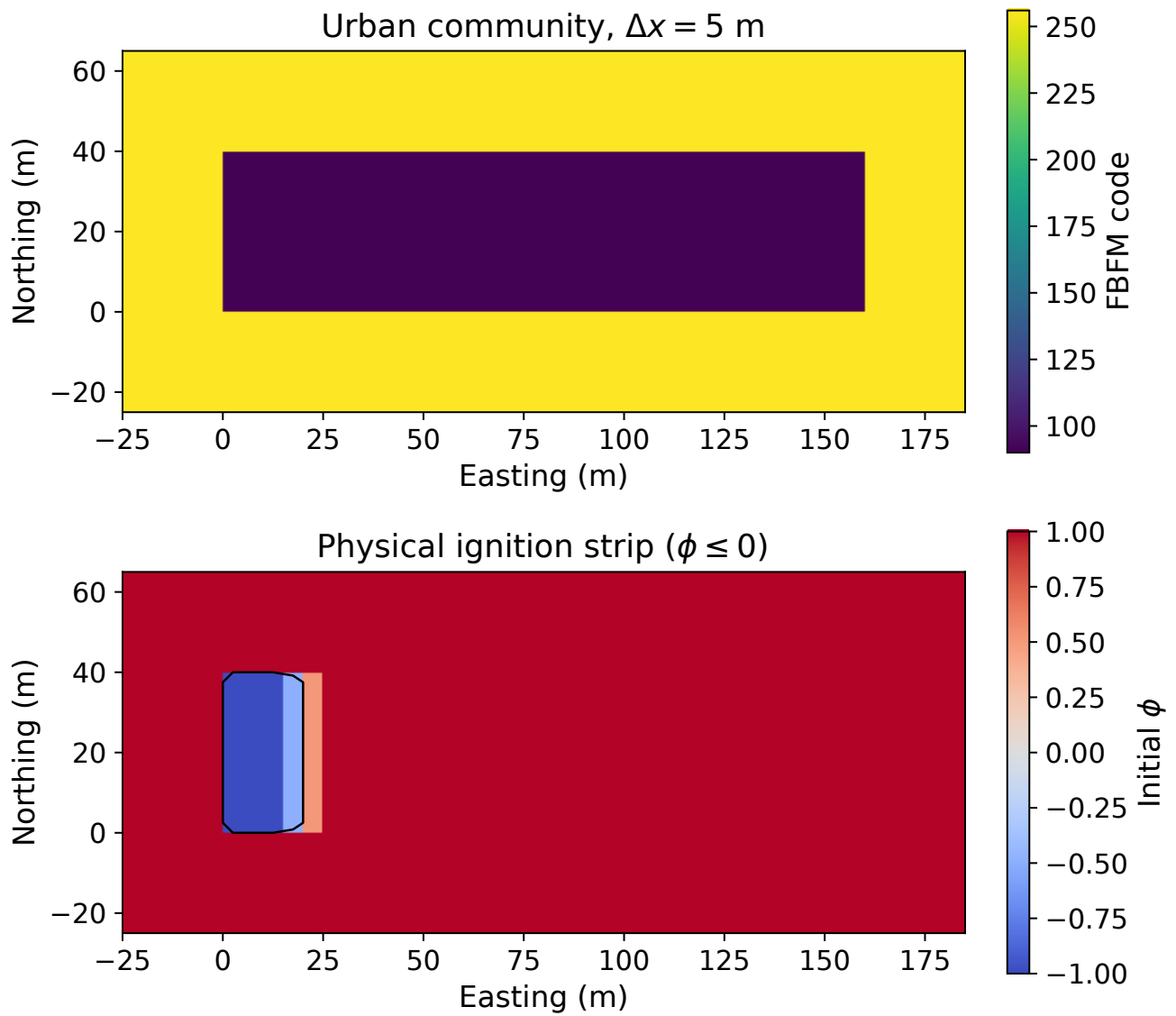


Figure 1: Mechanically rendered variants/dx_5p0m/data/inputs/fbfm.tif and phi.tif; this is the representative 5-m prepared grid, not a simulation result.

6 Actual simulation results

The workflow state is **COMPLETE**; the scientific decision is **FAIL**, with 4 of 4 required variants. Reason: One or more spatial-convergence acceptance components failed.. The executed binary SHA-256 is bc2d27483ed3cf409cea0a04fd879e290cfe5b57a0c84b215282ff9d18c458a9. For a completed record, `outputs/metrics.json` enumerates every exact selected raster path under `source_files` and every bound run receipt under `completion_receipt`; each receipt authenticates the executable, `stdout/stderr`, and complete output snapshot.

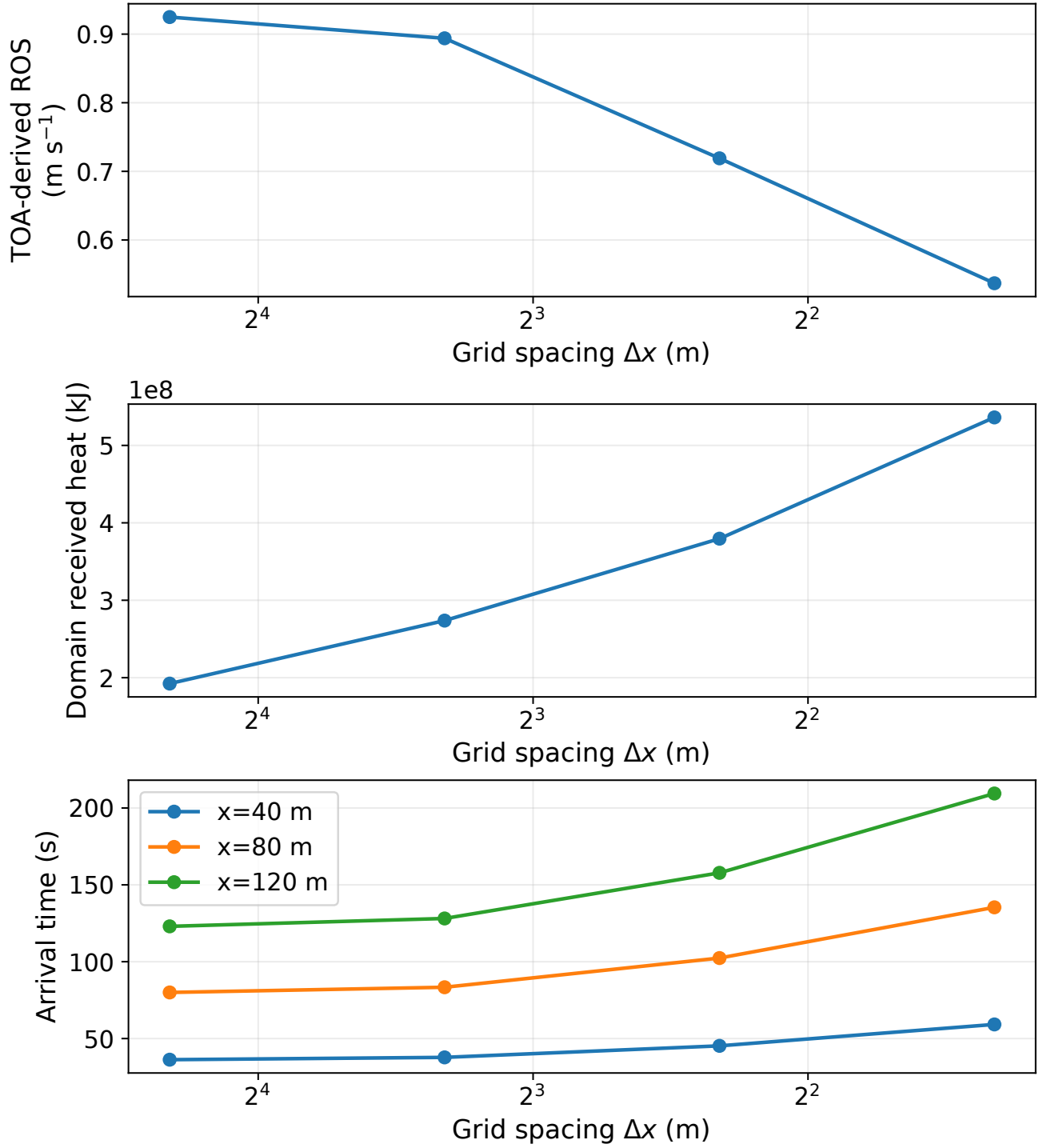


Figure 2: Exact- $t = 300$ s observables from the selected files recorded in `outputs/metrics.json`; the finest-pair limits are 5% for TOA/ROS and 10% for heat.

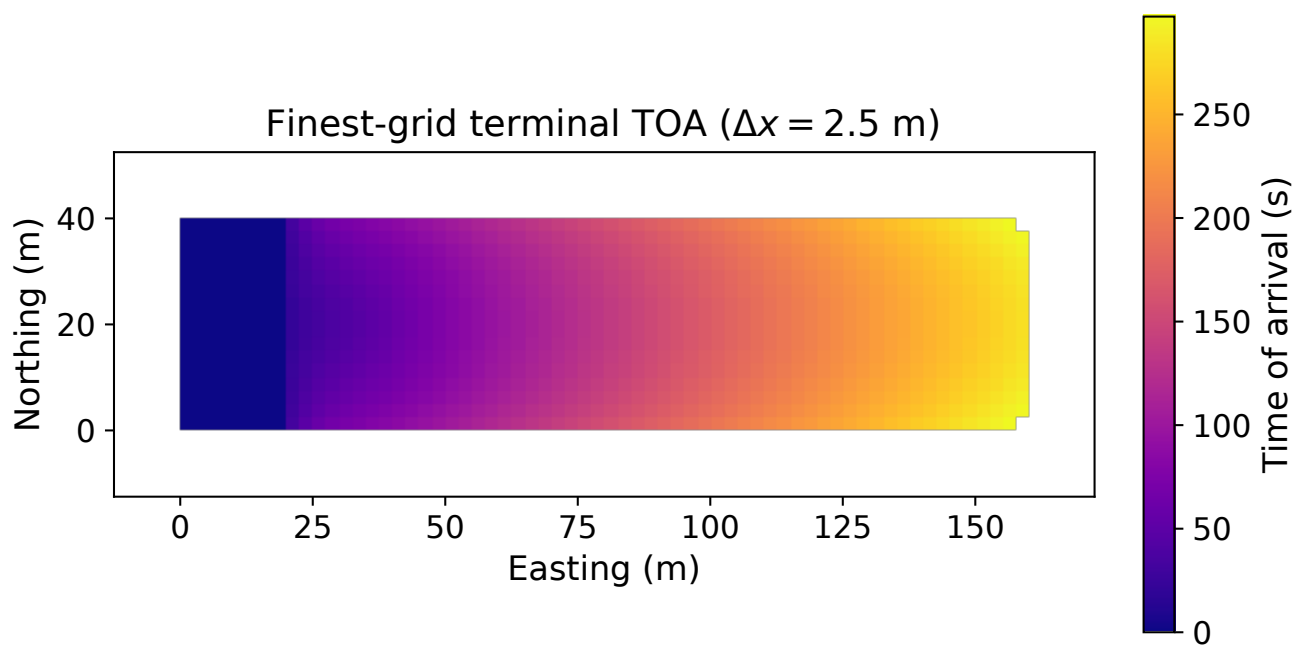


Figure 3: Finest-grid ($\Delta x = 2.5$ m) exact- $t = 300$ s TOA; the exact source raster path is recorded in `outputs/metrics.json`.

7 Calculated metrics and verification decision

Current-fingerprint output completeness — PASS

Criterion	4
Observed	<i>Units: variants.</i> 4

Finest-pair landmark TOA relative L2 change — FAIL

Criterion	≤ 0.05
Observed	<i>Units: fraction.</i> 0.245159

Finest-pair community ROS relative change — FAIL

Criterion	≤ 0.05
Observed	<i>Units: fraction.</i> 0.339079

Finest-pair total-heat relative change — FAIL

Criterion	≤ 0.1
Observed	<i>Units: fraction.</i> 0.292126

Landmark TOA observed spatial order — FAIL

Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-0.802156

Community ROS observed spatial order — FAIL

Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-0.0575922

Total-heat observed spatial order — FAIL

Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-0.565991

Minimum TOA-fit coefficient of determination — PASS

Criterion	≥ 0.95
Observed	0.999897

Maximum logged solver-step deviation — PASS

Criterion	≤ 0.051
Observed	<i>Units: s.</i> 0.05

The overall scientific decision is **FAIL**. Missing, stale, nonterminal, or malformed output is NOT EVALUABLE; NOT RUN is retained only as the workflow state and can never produce a pass.

8 Technical interpretation

All required grids have complete terminal rasters and the prescribed solver step sequences. The failed finest-pair and observed-order rows therefore represent sensitivity of the calculated solution, rather than absent files or an unintended temporal refinement. The fitted TOA profiles remain suitable for the reported community ROS, but that fitted ROS does not meet the registered grid-convergence criterion.

The source contains grid-dependent heat normalization, overlap/support rules, and an energy-dependent replacement of material FTP. These are plausible contributors to the observed sensitivity. The finite sequence identifies a failed convergence test; it does not uniquely assign causality or prove asymptotic divergence. The existing limits remain unchanged.

9 Scope and limitations

This is a numerical verification case for an idealized, fully urban, uniform community. It does not validate building-to-building fire behavior against experiments, represent explicit street gaps or variable structure footprints, exercise firebrands, or cover wildland–urban interfaces. CASE46 and CASE47 address transition pathways; CASE43–CASE45 isolate the component responses used here.

References

- [1] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. [doi:10.1071/WF24102](https://doi.org/10.1071/WF24102).